Here is the whole proof in miniature. Had no force acted at B, inertia (Law I) would carry the body straight on to c, with $Bc = AB$; then triangles SAB and SBc share the base SB and have equal heights, so $SAB = SBc$. But at B a single centripetal impulse toward S deflects the body, and by the parallelogram of motions (Law II, Cor. 1) it lands at C where $Cc \parallel SB$. Because Cc is parallel to SB, triangles SBc and SBC stand on the same base SB between the same parallels — so $SBc = SBC$, and therefore $SBC = SAB$. This is Newton's genius: an impulse pointed straight at S bends the body's *direction* but leaves the swept triangular area untouched.